



Dividing a quantity into a given ratio

Task A: Bar models

1) Use a bar model to answer the following:

a) Share 24 in the ratio 1 : 2 8 : 16

b) Share 48 in the ratio 3 : 1 36 : 12

c) Share 24 in the ratio 1 : 5 4 : 20

d) Share 48 in the ratio 7 : 1 42 : 6

e) Share 24 in the ratio 3 : 5 9 : 15

2) Use a bar model to answer the following:

a) Andeep, Sam and Sofia share 80 sweets in the ratio 3 : 8 : 5. How many sweets do they each get?

Andeep = 15, Sam = 40 and Sofia = 25

b) The perimeter of a triangle is 42.5 cm. The lengths of the sides are in the ratio of 7 : 6 : 4. How long is each side?

17.5 cm : 15 cm : 10 cm

c) The ratio of flour to butter to sugar in a cookie recipe is 3 : 2 : 1. You want to make a giant cookie weighing 510 grams. How much of each ingredient do you need?

Flour = 255 g, butter = 170 g and sugar = 85 g

Task B: Using efficient methods to share in a ratio

1) Use a ratio table to answer the following:

- a) Share 36 in the ratio 1 : 2 $12 : 24$
- b) Share 64 in the ratio 3 : 1 $48 : 16$
- c) Share 36 in the ratio 1 : 5 $6 : 30$
- d) Share 64 in the ratio 7 : 1 $56 : 8$
- e) Share 36 in the ratio 3 : 5 $13.5 : 22.5$

2) Use a ratio table to answer the following:

a) Andeep, Sam and Sofia share 240 sweets in the ratio 3 : 8 : 5. How many sweets do they each get?

Andeep = 45, Sam = 120 and Sofia = 75

b) The perimeter of a triangle is 107.1 cm. The lengths of the sides are in the ratio of 7 : 6 : 4. How long is each side?

44.1 cm : 37.8 cm : 25.2 cm

c) The ratio of flour to butter to sugar in a cookie recipe is 3 : 2 : 1. You want to make a giant cookie weighing 720 grams. How much of each ingredient do you need?

Flour = 360 g, butter = 240 g and sugar = 120 g

